Fist exam (A1)

Class: Bayesian Statistics
Instructor: Luiz Max Carvalho

- You have until April 13th 2026 at 14:20h Brasília time (GMT-3) to complete – and hand in – the exam;
- This is a take home exam. Feel free to consult with colleagues and classmates and to use AI assistants. **DO** however document everything you do: who gave you which ideas, what kinds of prompts you gave AI and how you interacted with it. The goal is to be principled, rigorous, transparent and awesome, just like Bayesian inference;
- The exam is worth $\min\{\text{your score}, 200\}/200$ marks.

1. The Magic Urn

In this simple exercise, we will warm up our muscles by looking at some simple consequences of exchangeability. In particular, we will be looking into the so-called Eggenberger-Pólya urn. Consider that, initially, an urn contains b_0 white balls and a_0 blue balls. At each iteration, a ball is drawn from the urn and then $s \geq 2$ balls of the same colour are returned to the urn. Thus, for example, if the first ball drawn is blue, then at the second iteration there will be b_0 white balls and s blue balls in the urn. Let X_i be the indicator that the i-th ball drawn is blue.

a) (20 marks) For each $n \in \mathbb{N}$, find the joint distribution of $(X_1, \ldots, X_n)$.

b) (20 marks) Show that $X_1, \ldots, X_n, \ldots$ is infinitely exchangeable.

c) (20 bonus marks) Use De Finetti's Theorem and the previous items to determine the distribution of $\lim_{n\to\infty} \bar{X}_n$.

2. The Ancestor's tail

In this exercise, we will make sure we know our Bayesian fundamentals by deducing the conjugate family for a simple model and deriving the Bayes estimator under different losses/utilities.

Consider that, given $\Theta = \theta$, $X_1, \ldots, X_n$ are i.i.d. with $X_i \sim \text{Uniform}(0, \theta)$.

(a) (10 marks) Deduce the conjugate prior in this case and find the expression for the posterior hyperparameters. Find the cumulative distribution function of the posterior distribution.

(b) (15 marks) Find the optimal estimator, δ_B, for θ under the losses:

- $L(\delta, \theta) = |\delta - \theta|$

- $L(\delta, \theta) = (\delta - \theta)^2$

(c) (15 marks) Construct the optimal credible interval, $[a, b]$, for θ under the utilities[1]:

- $U([a, b], \theta) = 100\mathbb{I}(\theta \in [a, b]) - 5(b - a)$
- $U([a, b], \theta) = -100\big((a - \theta)_+ + (\theta - b)_+\big) - 5(b - a)$
- $U([a, b], \theta) = -\frac{100}{b-a}\left(\theta - \frac{a+b}{2}\right)^2 - 5(b - a)$

(d) (10 marks) Is the posterior distribution of θ unimodal? Find a 95% highest posterior density interval (HPD) for θ.

3. It's only *natural*...

In this straightforward exercise we will investigate the Jeffreys's prior for the exponential family.

Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a regular natural exponential family with density

$$p(x \mid \theta) = h(x) \exp\{\theta^\top T(x) - A(\theta)\}.$$

a) (10 marks) **Deriving the Jeffreys's prior** Show that the Fisher information matrix is

$$I(\theta) = \nabla^2 A(\theta),$$

and hence derive Jeffreys's prior

$$\pi_J(\theta) \propto \sqrt{\det \nabla^2 A(\theta)}.$$

b) (20 marks) **Mean (re)parametrisation**

Let $\mu = \nabla A(\theta)$.

(i) Express the Fisher information in the μ-parametrisation.

(ii) Show that

$$\pi_J(\mu) \propto \sqrt{\det \mathrm{Var}_\theta[T(X)]^{-1}}.$$

c) (10 marks) **Factorisation**

Suppose

$$p(x \mid \theta) = \prod_{i=1}^{d} h_i(x) \exp\{\theta_i T_i(x) - A_i(\theta_i)\}.$$

(i) Show that the Fisher information matrix is diagonal.

(ii) Deduce that

$$\pi_J(\theta) \propto \prod_{i=1}^{d} \sqrt{A_i''(\theta_i)}.$$

(iii) Give an example of an exponential family where the Fisher information is not diagonal.

d) (10 marks) **(Im)propriety and boundary behaviour**

[1] Recall the relationship between utility and loss

(i) Show that Jeffreys's prior may be improper.

(ii) Analyse the behaviour of

$$\pi_J(\theta) \propto \sqrt{\det \nabla^2 A(\theta)}$$

near the boundary of Θ.

(iii) Consider:

- the Normal model with unknown variance,
- the Poisson model,

and determine whether Jeffreys's prior is proper in each case.

4. (Spatial) Smooth operator.

In this exercise we will explore the propriety of posterior distributions under the ICAR prior, a very common spatial smoothing model for areal data. The *intrinsic conditional autoregressive* (ICAR) model is a standard prior for spatially structured random effects. Because its precision matrix is rank-deficient, the ICAR specification yields an improper prior, and care is required to ensure the resulting posterior is proper.

a) (20 marks) **Rank deficiency of the ICAR precision matrix and its implications**.

Let $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_n)^\top$ be a vector of spatial random effects defined on a connected undirected graph $G = (V, E)$ with adjacency matrix $\mathbf{A}$ and diagonal degree matrix $\mathbf{D}$, where $D_{ii} = \sum_j A_{ij}$. The ICAR prior is specified up to a proportionality constant by

$$p(\boldsymbol{\phi} \mid \tau) \propto \ \tau^{(n-1)/2} \exp\Big\{-\tfrac{\tau}{2}\, \boldsymbol{\phi}^\top \mathbf{Q}\, \boldsymbol{\phi}\Big\}, \qquad \mathbf{Q} \coloneqq \mathbf{D} - \mathbf{A}. \tag{1}$$

(i) Show that $\mathbf{Q}$ is positive semi-definite with exactly one zero eigenvalue when G is connected. Identify the corresponding eigenvector explicitly.

(ii) Hence show that (1) does *not* define a proper probability distribution over $\mathbb{R}^n$.

(iii) State the identifiability constraint on $\boldsymbol{\phi}$ most commonly imposed in practice and explain its effect on the rank of $\mathbf{Q}$.

Hint. Use the graph-Laplacian characterisation $\boldsymbol{\phi}^\top \mathbf{Q} \boldsymbol{\phi} = \sum_{\{i,j\} \in E} (\phi_i - \phi_j)^2$ and the relationship between the null space of $\mathbf{Q}$ and constant vectors on a connected graph.

b) (30 marks) **Sufficient conditions for posterior propriety in a Poisson log-linear model**.

Consider the following hierarchical model for areal count data:

$$\begin{aligned} Y_i \mid \mu_i &\sim \text{Poisson}(E_i\, \mu_i), \\ \log \mu_i &= \boldsymbol{x}_i^\top \boldsymbol{\beta} + \phi_i, \\ \boldsymbol{\phi} \mid \tau &\sim \text{ICAR}(\tau), \\ \boldsymbol{\beta} &\sim \pi_B(\boldsymbol{\beta}), \\ \tau &\sim \pi_T(\tau). \end{aligned} \tag{2}$$

(i) State precisely what it means for the joint posterior $p(\boldsymbol{\beta}, \boldsymbol{\phi}, \tau \mid \mathbf{y})$ to be proper.

(ii) Prove that, under an improper prior $\pi_B(\boldsymbol{\beta}) \propto 1$ on $\boldsymbol{\beta} \in \mathbb{R}^p$ and the sum-to-zero constraint $\mathbf{1}^\top \boldsymbol{\phi} = 0$, the posterior is proper provided the likelihood is not constant as a function of $\boldsymbol{\beta}$ and the prior $p(\tau)$ is proper. State any additional conditions on the design matrix $\boldsymbol{X}$ that are required.

(iii) Explain why including an overall intercept in the linear predictor alongside the ICAR term creates an identifiability problem, and describe how it can resolved in practice.

Hint. Integrate out $\boldsymbol{\phi}$ using the rank-$(n-1)$ restricted Gaussian and analyse the resulting marginal likelihood in $\boldsymbol{\beta}$. Invoke the Dickey–Lindley condition to establish propriety of the marginal posterior of $\boldsymbol{\beta}$ – see e (u).

c) (40 marks) **Extension to the BYM2 reparameterisation and robustness of propriety**

The Besag–York–Mollié (BYM) s (e) is a staple of spatial statistics. It separates variation into two parts: one that captures similarities between neighbouring regions (spatial structure) and another that represents random noise or unexplained differences; by combining these two components, the model provides smoother and more realistic estimates of underlying risk patterns while still allowing for local irregularities.

The BYM2 model of e (i) is a reparametrisation of the BYM which improves interpretabilty. It decomposes the latent field as

$$\psi_i \;=\; \frac{1}{\sqrt{\tau}}\Big[\sqrt{\rho}\,\phi_i^* + \sqrt{1-\rho}\,\varepsilon_i\Big], \tag{3}$$

where $\boldsymbol{\phi}^*$ is the ICAR component scaled to have unit generalised variance, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ is an unstructured component independent of $\boldsymbol{\phi}^*$, and $\rho \in [0,1]$ controls the proportion of spatial variation.

(i) Show that for any $\rho \in (0,1]$ the joint prior on $\boldsymbol{\psi}$ is proper even without the sum-to-zero constraint, and characterise the boundary behaviour as $\rho \to 0$ and $\rho \to 1$.

(ii) Prove that, with a proper prior on (τ, ρ), the posterior under the Poisson likelihood (2) and a flat (improper) prior on $\boldsymbol{\beta}$ is proper under the same design-matrix conditions as in item (b).

(iii) Contrast the proof strategy with that of the pure ICAR case and discuss what role the unstructured component $\boldsymbol{\varepsilon}$ plays in regularising the posterior.

References

Besag, J., York, J., and Mollié, A. (1991). Bayesian image restoration, with two applications in spatial statistics. *Annals of the institute of statistical mathematics*, 43(1):1–20.

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